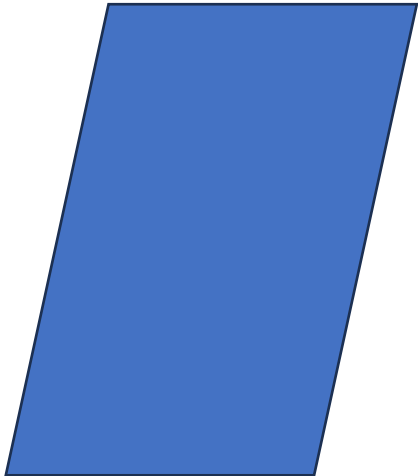
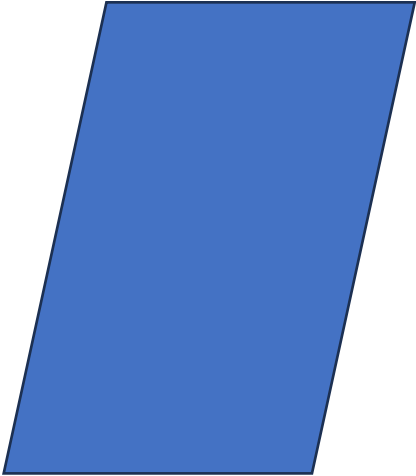
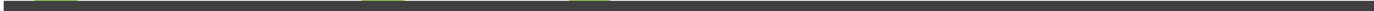




Core Architecture Principles



Learning Objectives

Understand the key principles that guide the design of effective, reliable, and scalable information systems.



Introduction:



Core Architecture Principles are the fundamental guidelines used in designing and managing systems—especially information systems—to ensure they are efficient, scalable, secure, and maintainable

They ensure that systems are:

- ☐ Efficient
- ☐ Secure
- ☐ Scalable
- ☐ Maintainable

Without these principles, systems may become:

- ☐ Difficult to manage
- ☐ Prone to failure
- ☐ Hard to integrate

Key Core Architecture Principles



MODULARITY:

Means dividing a system into smaller, independent parts called modules.

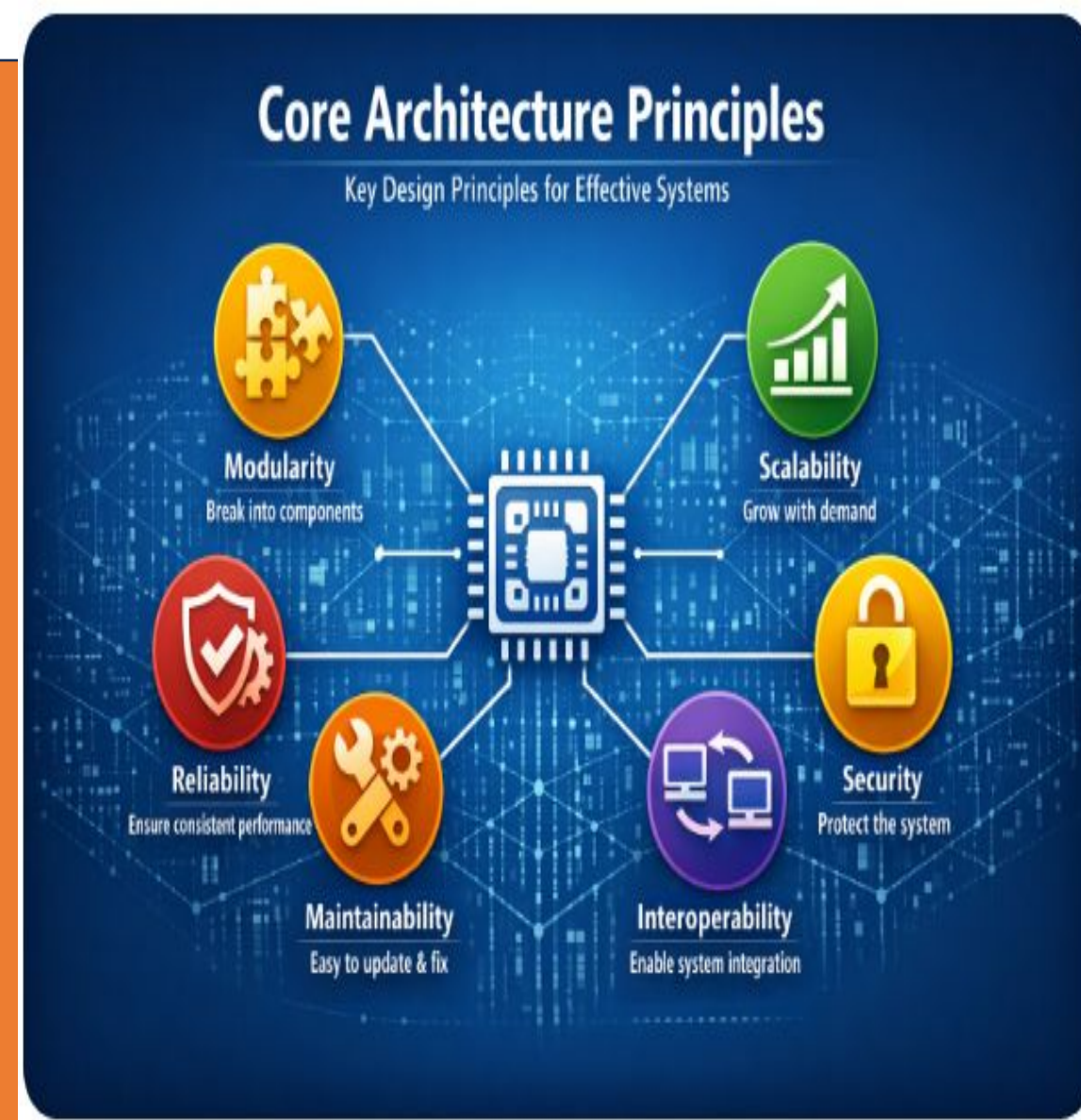
Why Modularity is Important

- ☐ Simplifies management
- ☐ Improves maintainability
- ☐ Supports scalability
- ☐ Enhances reusability
- ☐ Enables parallel development

Example

In a Student Information System:

- ❖ *Enrollment Module*
- ❖ *Grading Module*
- ❖ *Attendance Module*



Key Core Architecture Principles



SCALABILITY:

Ability of a system to handle growth in users, data, or workload..

Types of Scalability:

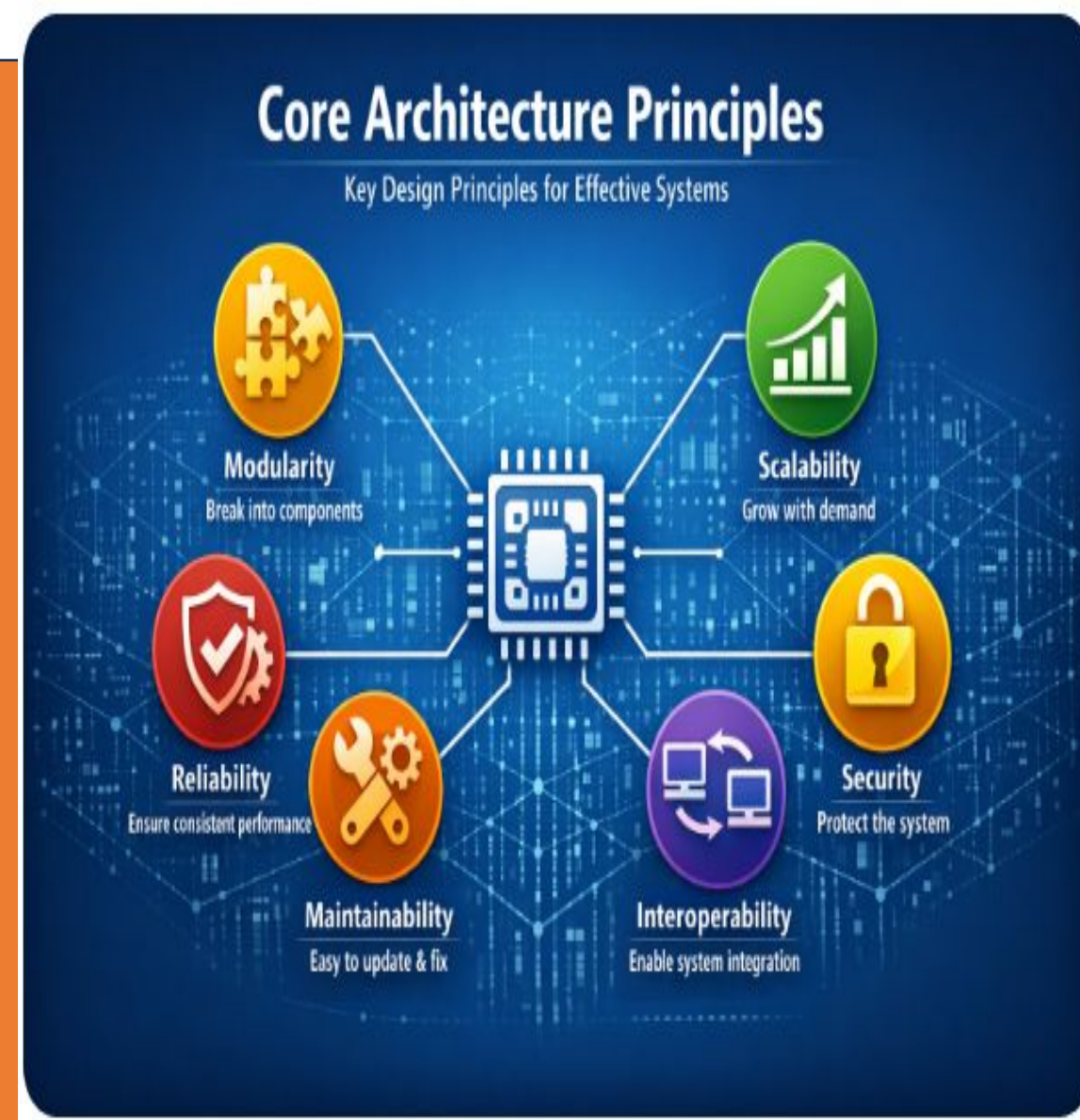
- ☐ Vertical Scalability (Scaling Up)
- ☐ Horizontal Scalability (Scaling Out)

Why Scalability is Important

- ☐ Supports organizational growth
- ☐ Prevents performance bottlenecks
- ☐ Future-proof system design

Example : An online system adds:

- ❖ *More servers*
 - ❖ *More storage*
 - ❖ *Cloud resources*
- as the number of users increases



Key Core Architecture Principles



RELIABILITY

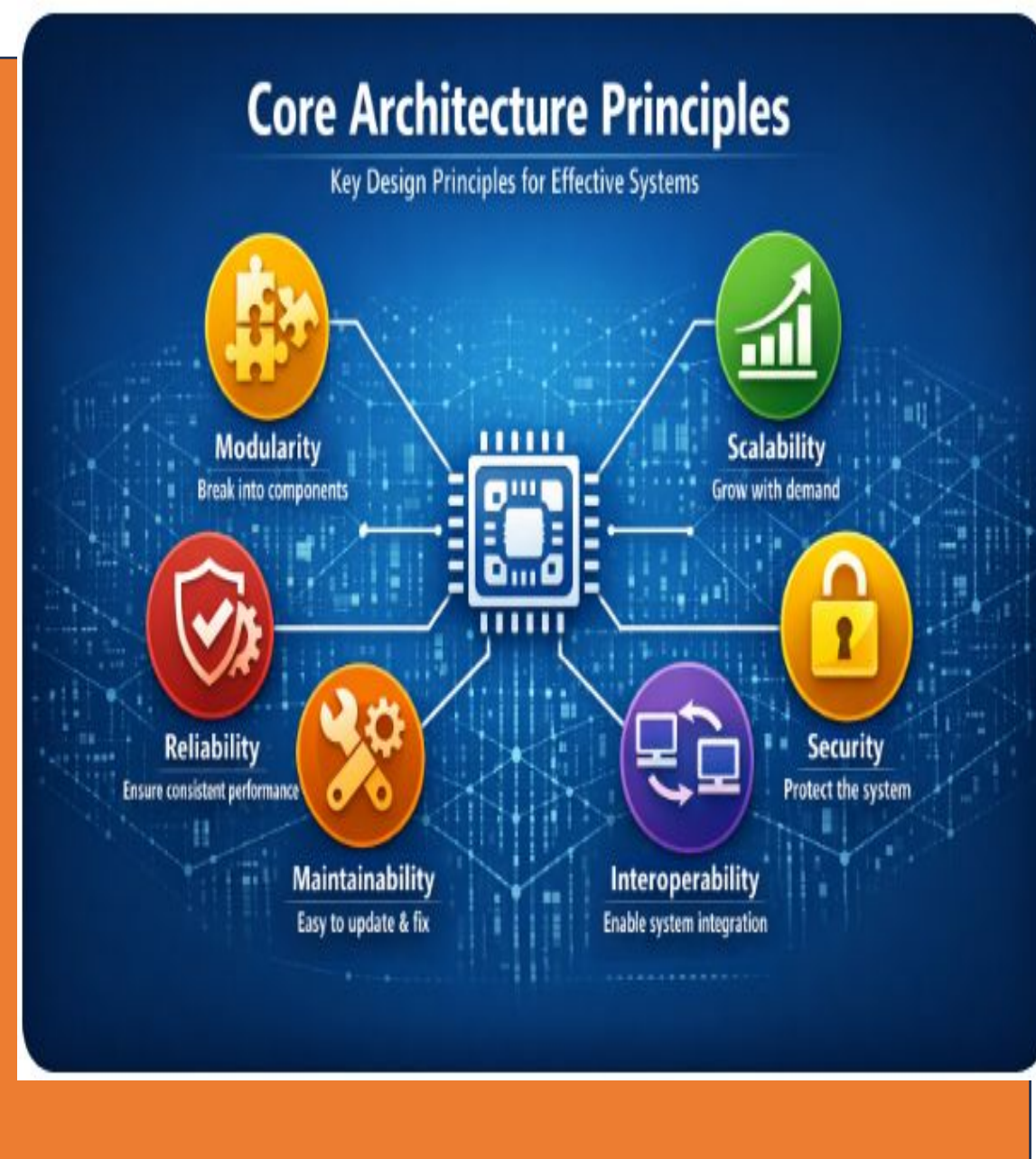
Ensures that the system consistently works without failure.

Why Scalability is Important

- ☐ Ensures continuous operation.
- ☐ Reduces downtime.
- ☐ Builds user trust.

Example :

- ❖ *Backup systems*
- ❖ *Failovers servers*
- ❖ *Error recovery mechanism*



Key Core Architecture Principles

Core Architecture Principles

Key Design Principles for Effective Systems



MAINTENABILITY

Refers to how easy it is to update, fix, or improve a system

Why Maintainability is Important

- ☐ Lower maintenance cost
- ☐ Faster updates
- ☐ Easier troubleshooting.

Example :

- ❖ Updating only one module (e.g., grading system)
- ❖ Fixing bugs without shutting down entire system

Key Core Architecture Principles



SECURITY

Ensures protection of system data and resources from unauthorized access.

Why Security is Important

- ☐ Protects sensitive data
- ☐ Prevents cyber attacks
- ☐ Ensures compliance with regulations

Example :

- ❖ User authentication
- ❖ Data encryption
- ❖ Access control

Key Core Architecture Principles

Core Architecture Principles

Key Design Principles for Effective Systems



INTEROPERABILITY

The ability of systems to communicate and work together.

Why Interoperability is Important

- ☐ Enables system integration
- ☐ Reduces data silos
- ☐ Improves organizational efficiency

Example :

- ❖ Integration via APIs
- ❖ Data exchange between systems
- ❖ Linking systems across departments

Key Core Architecture Principles

Core Architecture Principles

Key Design Principles for Effective Systems



In Summary:

PRINCIPLE	PURPOSE
Modularity	Break system into manageable parts
Scalability	Handle growth
Reliability	Ensure continuous operation
Maintainability	Ease of updates and fixes
Security	Protect system and data
Interoperability	Enable system communication
Modularity	Break system into manageable parts

Thank You